

Labeling the 45k dataset — MedGemma vs Mistral

We labelled the full **45,545-report** clinical EEG dataset (processed_reports) with our best pipeline, **MedGemma-27B v5g** (prompt v5 + grammar-enforced consistency, Q2_K, temperature 0). The paper’s **Mistral-7B** pipeline had already labelled the **same 45,545 reports** (released in the project database), so we have **two independent labelings of the whole set** and can compare them directly — and, on the reports that also carry human ground truth, see which labeler is more accurate.

Bottom line: the two models **agree on ~79% of reports** (all five present/absent calls) and 90-98% per category. Where they disagree it is mostly the *slowing* classes — and there **MedGemma is the more reliable one**: on the 2,493 reports with human labels, MedGemma matches the annotator **87.1%** of the time vs Mistral’s **75.0%**.

How much the two labelings agree (all 45,545)

Category	MedGemma “present”	Mistral “present”	Agreement
Abnormality	42.6%	40.8%	93.4%
Focal Epi	7.3%	8.1%	98.2%
Gen Epi	6.0%	6.3%	98.0%
Focal Non-epi	23.2%	22.0%	89.9%
Gen Non-epi	20.8%	17.3%	91.2%

Whole-report agreement (present/absent, all 5): 79.1%. On the exact 1-4 confidence level the two agree only 42.6% — expected, since the models express confidence differently (Mistral’s certainty calibration is the weaker part of that pipeline).

Where they differ: almost entirely the **non-epileptiform (slowing)** classes. The two are near-identical on the rare epileptiform findings (98% agreement) but MedGemma flags **Gen Non-epi** noticeably more often (20.8% vs 17.3%) — the diffuse-slowng / encephalopathy class that Mistral tends to under-call.

Which one is right? (accuracy vs the human annotator)

2,493 of the 45,545 reports also have human ground truth (annotator LD, from the Zoe/Maria annotation sets). On those we can score both labelings against the human:

Category	MedGemma	Mistral
Abnormality	95.3	95.1
Focal Epi	86.8	83.4
Gen Epi	88.2	73.4
Focal Non-epi	88.2	75.3
Gen Non-epi	86.4	75.9
Whole report	87.1%	75.0%

(Core F1 vs LD; whole report = share with all five present/absent calls correct.)

The two are tied on the easy overall abnormal/normal call (95 vs 95), but on every harder class **MedGemma is well ahead** — decisively on **Gen Epi (88 vs 73)**, **Focal Non-epi (88 vs 75)** and **Gen Non-epi (86 vs 76)**. This is exactly where the two labelings disagreed on the full set — so the disagreements resolve in MedGemma’s favour.

Takeaways

- **Ship the MedGemma labeling** as the primary one for the 45k set — it agrees with the human annotator 12 points more often than Mistral (87% vs 75%), driven by the harder slowing/generalized

Accuracy on the annotated reports — MedGemma vs Mistral

Core F1 vs human ground truth (LD) · n=2493 annotated of 45,545

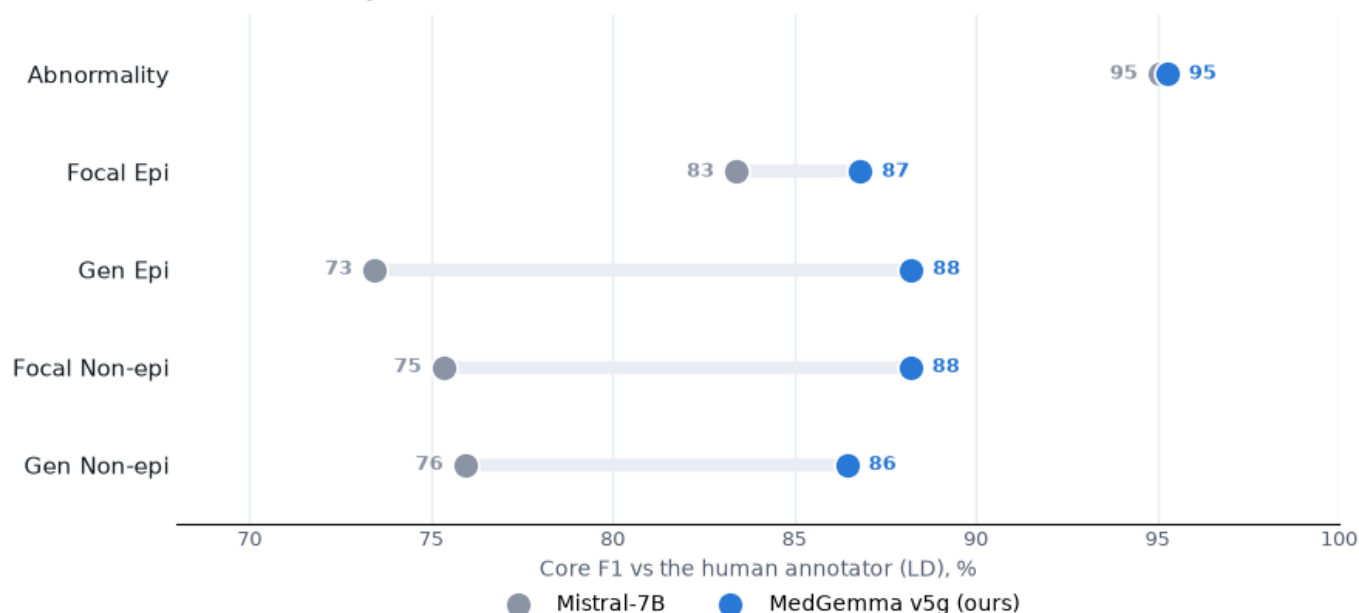


Figure 1: Accuracy vs the human annotator

classes.

- **Mistral is a useful independent cross-check.** The ~21% of reports where the two disagree — concentrated in the slowing classes — are a natural shortlist for human spot-review.
- **Confidence levels are not comparable across the two** (42.6% exact agreement); if the downstream use needs the 1–4 level, use MedGemma’s (its certainty was validated far closer to the human than Mistral’s).

Reproduce

```
python -m analysis.labels_vs_mistral # tables above + the chart
```

MedGemma labels: results/labels/*.json (produced by slurm/label_reports.sbatch, one JSON per 2,000-report chunk). Mistral labels: the released classifications table. Human ground truth: the Zoe/Maria LD annotation databases. Labels store only the five scores and confidences per hashed report id — never the report text.